Lecture-1

Laplace Transformation

Definition: Let the function $f(t)$ be defined for all positive values of t, then multiply $f(t)$bye^{-st} and integrate it with respect to t from zero to infinity. If the resulting integral exists (i.e., has some finite value), it is a function of s, s may be real or complex, say $F(s)$.

$$F(s) = \int_0^\infty e^{-st} f(t)dt.$$

This function $F(s)$ of variable s is called Laplace Transformation of the original function $f(t)$ and will be denoted by $\mathcal{L}\{f(t)\}$, where $\mathcal{L}$ denotes the Laplace transform operator. Thus

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)dt.$$

The original function $f(t)$ is called the inverse transform or inverse of $F(s)$ and will be denoted by$\mathcal{L}^{-1}\{F(s)\}$.

$$\therefore f(t) = \mathcal{L}^{-1}\{F(s)\}.$$

Important formulae:

1. $\mathcal{L}\{c\} = \frac{c}{s}$, c is any constant, $(s > 0)$	**Some important formulae**
2. $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$, when $n = 0,1,2,3,\ldots$	$\sin at = \frac{e^{iat} - e^{-iat}}{2i}$
3. $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, (s > a)$	$\cos at = \frac{e^{iat} + e^{-iat}}{2}$
4. $\mathcal{L}\{\cosh at\} = \frac{s}{s^2-a^2}, \lvert s\rvert > a$	$\sinh at = \frac{e^{at} - e^{-at}}{2}$
5. $\mathcal{L}\{\sinh at\} = \frac{a}{s^2-a^2}, \lvert s\rvert > a$	$\cosh at = \frac{e^{at} + e^{-at}}{2}$
6. $\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}, \lvert s\rvert > a$	$\int e^{at} \sin bt \ dt = \frac{e^{at}(a\sin bt - b\cos bt)}{a^2+b^2}$
7. $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}, \lvert s\rvert > a$	$\int e^{at} \cos bt \ dt = \frac{e^{at}(a\cos bt + b\sin bt)}{a^2+b^2}$

Properties of Laplace transformation:

1. $\mathcal{L}\{af_1(t) + bf_2(t)\} = a\mathcal{L}\{f_1(t)\} + b\mathcal{L}\{f_2(t)\}$**(linearity)**, where a &b are constants.

2. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$**(first shifting or translation)**

3. If $\mathcal{L}\{f(t)\} = F(s)$, then $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}[F(s)]$**(multiplication by t^n)**

Proof of some selective formulae:

1. $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, (s > a)$

Proof: From the definition of Laplace transformation we know that,

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)dt$$

$$\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-st}e^{at}dt = \lim_{p\to\infty}\int_0^p e^{-(s-a)t}dt = \lim_{p\to\infty}\left[\frac{e^{-(s-a)t}}{-(s-a)}\right]\Bigg|_0^p = \lim_{p\to\infty}\frac{1-e^{-(s-a)p}}{(s-a)}$$

$= \frac{1}{s-a}$ if $s > a$.

2. $\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}$ if $|s| > a$ and 3. $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$ if $|s| > a$.

Proof: $\mathcal{L}\{e^{iat}\} = \frac{1}{s-ia} = \frac{s+ia}{s^2+a^2} = \frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}$ - - - - - (1)

But, $e^{iat} = \cos at + i\sin at$

So, $\mathcal{L}\{e^{iat}\} = \int_0^\infty e^{-st}e^{iat}dt = \int_0^\infty e^{-st}(\cos at + i\sin at)dt$

$$= \int_0^\infty e^{-st}\cos at\, dt + i\int_0^\infty e^{-st}\sin at\, dt$$

$= \mathcal{L}\{\cos at\} + i\ \mathcal{L}\{\sin at\}$ - - - - - - (2)

Comparing (1) and (2), we have on equating real and imaginary parts,

$\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}$ if $|s| > a$ and $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$ if $|s| > a$.

Some workout examples on Laplace transformation:

Example: 1	**Example: 2**
$\mathcal{L}\{(t^2+1)^2\}$ $= \mathcal{L}\{t^4 + 2t^2 + 1\}$ $= \mathcal{L}\{t^4\} + 2\mathcal{L}\{t^2\} + \mathcal{L}\{1\}$ $= \frac{4!}{s^{4+1}} + 2\frac{2!}{s^{2+1}} + \frac{1}{s}$ $= \frac{24}{s^5} + \frac{4}{s^3} + \frac{1}{s}$	$\mathcal{L}\{e^{-3t} + 5\cosh t\}$ $= \mathcal{L}\{e^{-3t}\} + 5\,\mathcal{L}\{\cosh\ t\}$ $= \frac{1}{s-(-3)} + 5\frac{s}{s^2-1^2}$ $= \frac{1}{s+3} + \frac{5s}{s^2-1}$

Example: 3

$\mathcal{L}\{(\sin t - \cos t)^2 + \cos^2 3t\}$

$= \mathcal{L}\left\{\sin^2 t - 2\sin t\cos t + \cos^2 t + \frac{1}{2}(1 + \cos 6t)\right\}$

$= \mathcal{L}\left\{1 - \sin 2t + \frac{1}{2} + \frac{1}{2}\cos 6t)\right\}$

$= \mathcal{L}\left\{\frac{3}{2}\right\} - \mathcal{L}\{\sin 2t\} + \frac{1}{2}\mathcal{L}\{\cos 6t\}$

$= \frac{3}{2}\frac{1}{s} - \frac{2}{s^2+2^2} + \frac{1}{2}\frac{s}{s^2+6^2}$

$= \frac{3}{2s} - \frac{2}{s^2+4} + \frac{1}{2}\left(\frac{s}{s^2+36}\right).$

Example: 4

$\mathcal{L}\{3 - \mathrm{e}^{-t}\cos 2t + t\ \sinh 3t\}$

$= \mathcal{L}\{3\} - \mathcal{L}\{\mathrm{e}^{-t}\cos 2t\} + \mathcal{L}\{t\sinh 3t\}$

$= 3\frac{1}{\mathrm{s}} - \frac{\mathrm{s}+1}{(\mathrm{s}+1)^2+2^2} + (-1)^1\frac{\mathrm{d}}{\mathrm{ds}}\left(\frac{3}{\mathrm{s}^2-3^2}\right)$

$= \frac{3}{\mathrm{s}} - \frac{\mathrm{s}+1}{\mathrm{s}^2+2\mathrm{s}+5} + \frac{3.2\mathrm{s}}{(\mathrm{s}^2-9)^2}$

$= \frac{3}{\mathrm{s}} - \frac{\mathrm{s}+1}{\mathrm{s}^2+2\mathrm{s}+5} + \frac{6\mathrm{s}}{(\mathrm{s}^2-9)^2}$

Example: 5

$\mathcal{L}\{t\cos t - \mathrm{e}^{-3t}\sin 2t + t^7\mathrm{e}^{5t}\}$

$= (-1)^1\frac{\mathrm{d}}{\mathrm{ds}}\frac{\mathrm{s}}{\mathrm{s}^2+1} - \frac{2}{(\mathrm{s}+3)^2+2^2} + \frac{7!}{(\mathrm{s}-5)^8}$

$= -\frac{1-\mathrm{s}^2}{(\mathrm{s}^2+1)^2} - \frac{2}{(\mathrm{s}+3)^2+4} + \frac{5040}{(\mathrm{s}-5)^8}$

Example: 6

$\mathcal{L}\{\mathrm{e}^{2t}t + \mathrm{e}^t\cosh 3t\}$

$= \frac{1}{(\mathrm{s}-2)^2} + \frac{\mathrm{s}-1}{(\mathrm{s}-1)^2-3^2}$

$= \frac{1}{(\mathrm{s}-2)^2} + \frac{\mathrm{s}-1}{\mathrm{s}^2-2\mathrm{s}-8}.$

Example: 7

$\mathcal{L}\{t\sin at\}$

$= \mathcal{L}\left\{t\frac{e^{iat}-e^{-iat}}{2i}\right\}$

$= \frac{1}{2i}\mathcal{L}\{te^{iat} - te^{-iat}\}$

$= \frac{1}{2i}\left[\frac{1!}{(s-ia)^2} - \frac{1!}{(s+ia)^2}\right]$(first Shifting Property)

$= \frac{2as}{(s^2+a^2)^2}$

Or

$\mathcal{L}\{t\sin at\}$

$= (-1)^1\frac{d}{ds}\left(\frac{a}{s^2+a^2}\right)$

$= -\frac{(s^2+a^2).0 - a.2s}{(s^2+a^2)^2}$

$= -\frac{-2as}{(s^2+a^2)^2}$

$= \frac{2as}{(s^2+a^2)^2}$

Example: 8

$\mathcal{L}\{t\cos at\}$

$= \mathcal{L}\left\{t\frac{e^{iat}+e^{-iat}}{2}\right\}$

$= \frac{1}{2}\mathcal{L}\{e^{iat}t + e^{-iat}t\}$

$= \frac{1}{2}\left[\frac{1!}{(s-ia)^2} + \frac{1!}{(s+ia)^2}\right]$(first shifting property)

$= \frac{a^2-s^2}{(s^2+a^2)^2}$

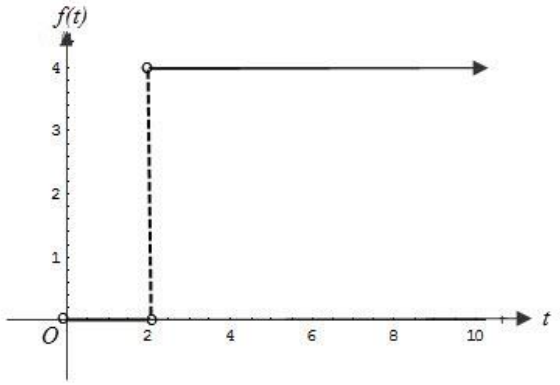
Or

$\mathcal{L}\{t\cos at\}$

$= (-1)^1\frac{d}{ds}\left(\frac{s}{s^2+a^2}\right)$

$= \frac{(s^2+a^2).1 - s.2s}{(s^2+a^2)^2}$

$= \frac{a^2-s^2}{(s^2+a^2)^2}$

Piecewise function	
Example: Sketch $f(t)=\begin{cases}0, & 0<t<2\\ 4, & t>2\end{cases}$ and find $\mathcal{L}\{f(t)\}$. **Solution:** $$\mathcal{L}\{f(t)\}=\int_0^\infty e^{-st}f(t)dt$$ $$=\int_0^2 e^{-st}.0\,dt+\int_2^\infty e^{-st}\,4\,dt$$ $$=4\left[\frac{e^{-st}}{-s}\right]\Big|_2^\infty=4\frac{e^{-2s}}{s}.$$	

Exercises set 1.1

Advanced Engineering Mathematics- Erwin Kreyszig.(10th Edition) , JOHN, WILEY & SONS, INC

Page no: 210, Problem no: (1- 6), 9, 11, 12.

Find the Laplace Transformations of the following functions. Also sketch the indicated functions. Also verify the answers using MATLAB command:

	$f(t)$	Answer $F(s)$	..		$f(t)$	Answer
1.	$t^2, e^{3t}, e^{-t},$ $\sin 3t, \cos 2t,$ $(e^t-1)^2$.			10.	$t\cos at$	$\frac{s^2-a^2}{(s^2+a^2)^2}$
2.	$\sin^2 2t$	$\frac{1}{2s}-\frac{s}{2(s^2+16)}$		11.	$4\cos 3t-2\sin 3t+t^3$	$\frac{4s}{s^2+9}-\frac{6}{s^2+9}+\frac{6}{s^4}$
3.	t^3e^{2t}	$\frac{6}{(s-2)^4}$		12.	$\sin 2t+5\cos t$	$\frac{2}{s^2+4}+\frac{5s}{s^2+1}$
4.	$t\sin 5t$	$\frac{10s}{(s^2+25)^2}$		13.	$\frac{1}{4}\sin 4t-4\cos\left(\frac{t}{2}\right)$	$\frac{1}{s^2+16}-\frac{16s}{4s^2+1}$
5.	$t\cos 8t$	$\frac{s^2-64}{(s^2+64)^2}$		14.	$e^{-t}\sin 5t+4e^{-t}\cos 5t$	$\frac{9+4s}{(s+1)^2+25}$
6.	$e^{-2t}\cos 5t$	$\frac{s+2}{(s+2)^2+25}$		15.	$e^{-t}\cos^2 t$	$\frac{1}{2s+2}+\frac{s+1}{2s^2+4s+10}$
7.	$2t(2t^4e^{-3t}+1)$	$\frac{480}{(s+3)^6}+\frac{2}{s^2}$			**Piecewise function**	
8.	$t\sinh 3t$	$\frac{6s}{(s^2-3^2)^2}$		16.	Sketch $f(t)=\begin{cases}t^2, 0\le t\le 2\\ 0, t>2\end{cases}$	$-\frac{4}{s}e^{-2s}-\frac{4}{s^2}e^{-2s}-\frac{2}{s^3}e^{-2s}+\frac{2}{s^3}$.

9.	te^{7t}	$\frac{1}{(s-7)^2}$		17.	Sketch $f(t)=\begin{cases}3t, 0\le t\le 5\\ 1, t>5\end{cases}$	$\frac{3}{s^2}(1-e^{-5s})-\frac{14}{s}e^{-5s}$

18. Solve the following differential equations:
(i) $\dot{y}(t)=2; y(0)=3.$
(ii) $\dot{y}(t)=t; y(0)=2.$
(iii) $\dot{y}(t)=e^t; y(0)=0.$
(iv) $\dot{y}(t)=\sin t\,; y(0)=0.$

MATLAB command to find Laplace transformation:

$f(t)=-1.25+3.5te^{-2t}+1.25e^{-2t}$

```
>>syms t s

>> f=-1.25+3.5*t*exp(-2*t)+1.25*exp(-2*t);

>>laplace(f, t, s)

ans =

5/(4*(s + 2)) + 7/(2*(s + 2)^2) - 5/(4*s)
```

Laplace transformation of periodic function

If $f(t)$, defined for all positive t, is a periodic function with period $P>0$, i. e. $f(t+nP)=f(t), \forall\, t$ in the domain of $f(t), n=0,1,2,3$ .... Then the Laplace transformation of $f(t)$ is,

$$\mathcal{L}\{f(t)\}=\frac{1}{1-e^{-sP}}\int_0^P e^{-st}f(t)dt.$$

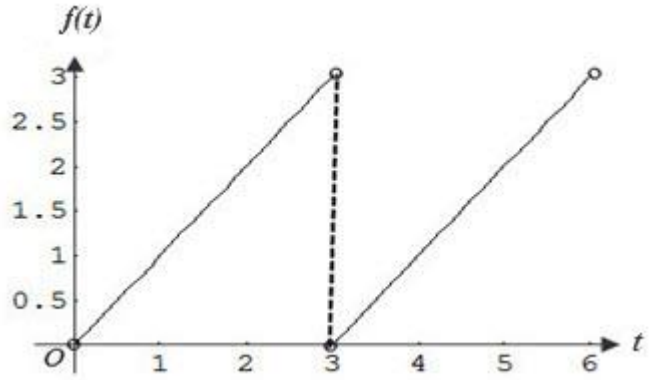
Example:1

Let $f(t) = t,\ 0 < t < 3$ is a periodic function of period 3. Then

(i) sketch *f(t)*and

(ii) find the Laplace transformation of the above function .

Solution: (i)



Solution: (ii)

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-SP}}\int_0^T e^{-st}f(t)dt$$

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-3s}}\int_0^3 e^{-st}tdt$$

$$= \frac{1}{1-e^{-3s}}\left[t\frac{e^{-st}}{-s} - \frac{e^{-st}}{s^2}\right]\Bigg|_0^3$$

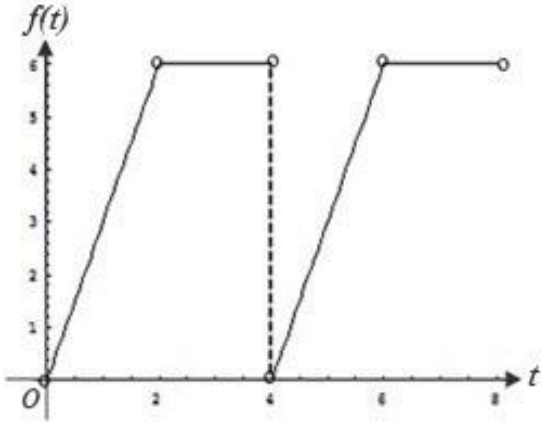
$$= \frac{1}{1-e^{-3s}}\left[\frac{e^{-3s}}{-s} - \frac{e^{-3s}}{s^2} + \frac{1}{s^2}\right]$$

Example: 2

Let $f(t) = \begin{cases} 3t, & 0 < t < 2 \\ 6, & 2 < t < 4 \end{cases}$ is a periodic function of period 4. Then

(i) sketch *f(t)*and(ii) find the Laplace transformation of the above function .

Solution: (i)



Solution: (ii)

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-4s}}\int_0^4 e^{-st}f(t)dt$$

$$= \frac{1}{1-e^{-4s}}\left[\int_0^2 e^{-st}3tdt + \int_2^4 e^{-st}6\,dt\right]$$

$$= \frac{1}{1-e^{-4s}}\left[3\left(t\frac{e^{-st}}{-s} - \frac{e^{-st}}{s^2}\right)\Big|_0^2 + 6\left(\frac{e^{-st}}{-s}\right)\Big|_2^4\right]$$

$$= \frac{1}{1-e^{-4s}}\frac{1}{s^2}(3 - 3e^{-2s} - 6se^{-4s})$$

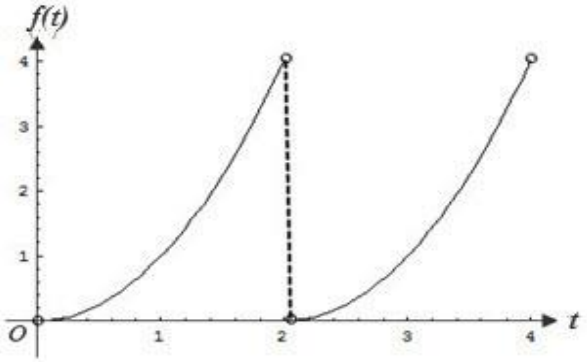
Example: 3

Let $f(t) = t^2,\ 0 < t < 2$ and $f(t+2) = f(t)$.Then

(i) sketch *f(t)* and

(ii) find the Laplace transformation of the above function .

Solution: (i)



Solution: (ii)

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2S}}\int_0^2 e^{-st}t^2dt$$

$$= \frac{1}{1-e^{-2S}}\left[t^2\frac{e^{-st}}{-s} - 2t\frac{e^{-st}}{s^2} + 2\frac{e^{-st}}{-s^3}\right]\Bigg|_0^2$$

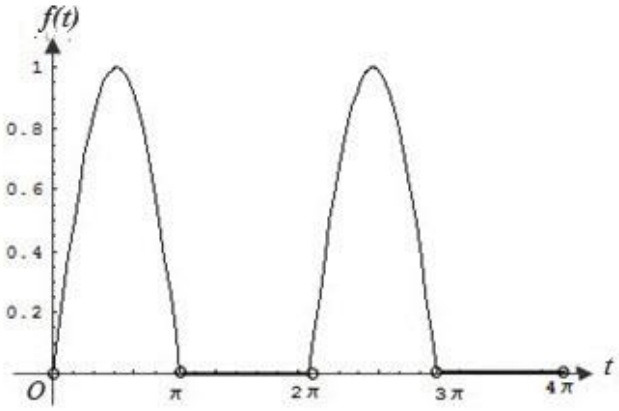
$$= \frac{1}{1-e^{-2S}}\left[-\frac{4}{s}e^{-2s} - \frac{4}{s^2}e^{-2s} - \frac{2}{s^3}e^{-2s} + \frac{2}{s^3}\right]$$

Example: 4

Let $f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 0, & \pi < t < 2\pi \end{cases}$ is a periodic function of period 2π. Then

(i) sketch *f(t)*and(ii) find the Laplace transformation of the above function .

Solution: (i)



Solution: (ii)

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-2\pi S}}\int_0^{2\pi} e^{-st}f(t)dt$$

$$= \frac{1}{1-e^{-2\pi S}}\left[\int_0^{\pi} e^{-st}\sin t\,dt + \int_{\pi}^{2\pi} e^{-st}.0\,dt\right]$$

$$= \frac{1}{1-e^{-2\pi S}}\left[\frac{e^{-st}(-s\sin t - \cos t)}{s^2+1}\right]\Bigg|_0^{\pi}$$

$$= \frac{1}{1-e^{-2\pi S}}\frac{1+e^{-\pi s}}{s^2+1}$$

Exercises set 1.2

Sketch *f(t)*and find the Laplace transformation of the following periodic functions.

1. $f(t) = e^{2t}, 0 < t < \pi, \text{period } T = \pi.$ Ans: $F(s) = \frac{e^{(2-s)\pi}-1}{(2-s)(1-e^{-\pi s})}$.

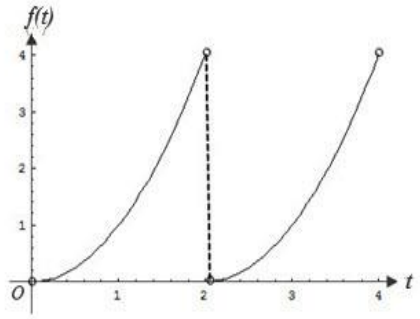
2. $f(t) = t, 0 < t < 1, f(t+1) = f(t)$. Ans: $F(s) = \frac{1}{1-e^{-s}}\left(-\frac{1}{s}e^{-s} - \frac{1}{s^2}e^{-s} + \frac{1}{s^2}\right)$.

3. $f(t) = \begin{cases} t, & 0 < t < 2 \\ 2, & 2 < t < 4 \end{cases}$, $f(t+4) = f(t)$. Ans: $f(s) = \frac{1-e^{-2s}-2se^{-4s}}{s^2\left(1-e^{-4s}\right)}$.

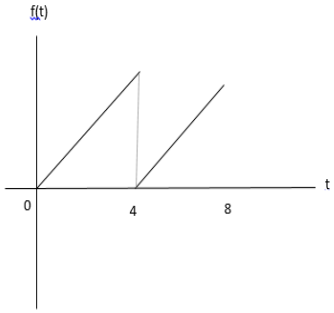
4. $f(t) = \begin{cases} t, & 0 < t < 2 \\ 2-t, & 2 < t < 6 \end{cases}$, $\text{period } T = 6$. Ans: $f(s) = \frac{1-2e^{-2s}-2se^{-2s}+4se^{-6s}+e^{-6s}}{\left(1-e^{-6s}\right)s^2}$.

6. Represent the following functions in terms of periodic function and find their Laplace transformations:

i)



ii)



The Unit step function (Heavisidefunction):

In engineering applications, we frequently encounter functions whose values change abruptly at specified values of time *t*. One common example is when a voltage is switched on or off in an electrical circuit at a specified value of time *t*.

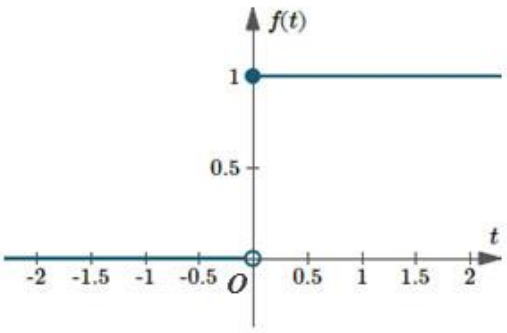
The value of *t* = 0 is usually taken as a convenient time to switch on or off the given voltage.

The switching process can be described mathematically by the function called the **Unit Step Function** (otherwise known as the **Heaviside function** after Oliver Heaviside).

The Unit step function or Heaviside's unit step function, $u(t)$,is defined as follows:

$$f(t) = u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases},$$

That is, *u* is a function of time *t*, and *u* has value **zero** when time is negative (before we flip the switch); and value **one** when time is positive or zero (from when we flip the switch).



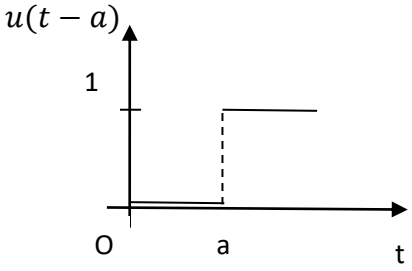
Graph of $f(t) = u(t)$, the unit step function.

Shifted (Right) Unit step function:

In many circuits, waveforms are applied at specified intervals other than t=0. Such a function may be described using the **shifted** (aka **delayed**) unit step function.

A function which has value0 up to the time $t = a$ and thereafter has value 1 is written:

$$u_a(t) = u(t-a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases},$$



Rectangular pulse:

A common situation in a circuit is for a voltage$V(t)$ to be applied at a particular time (say $t = a$) and removed later, at $t = b$ (say). We write such a situation using unit step functions as:

$$V(t) = u(t-a) - u(t-b)$$

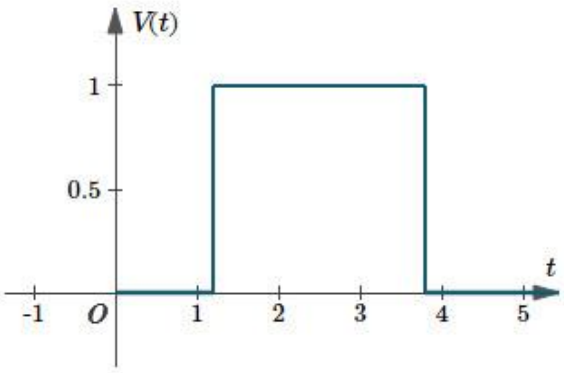
This voltage has strength 1, duration $(b-a)$.

Alternatively, $V(t)$ may be constructed using top hat function $u(t-a) - u(t-b)$ as follows:

$$V(t) = u(t-a) - u(t-b) = \begin{cases} 1, & a \leq t < b \\ 0, & \text{otherwise} \end{cases},$$

Example01 :

The graph of $V(t) = u(t-1.2) - u(t-3.8)$ is as follows. Here, the duration is 3.8−1.2=2.6 .



Graph of$V(t) = u(t - 1.2) - u(t - 3.8)$is an example of a rectangular pulse.

Example 02:
Write the following functions in terms of **unit step** function(s). Sketch each waveform.

(i) A 12-V source is switched on at t = 4 s,

(ii) $V(t) = \begin{cases} 1, & 0 < t < a \\ 0, & t > a \end{cases}$ and,

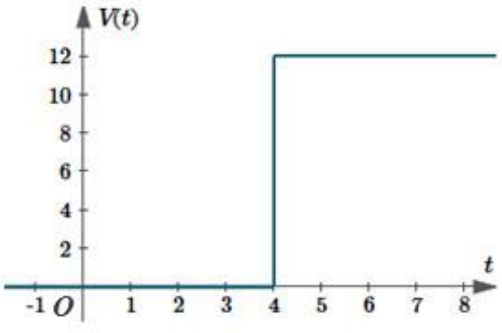
(iii) $V(t) = \begin{cases} 0, & t < 3 \\ 2t + 8, & 3 < t < 5 \\ 0, & t > 5 \end{cases}$.

Solution:

(i) Since the voltage is turned on at t = 4, we need to use $u(t - 4)$.We multiply by 12 since that is the voltage.

We write the function as follows: $V(t) = 12 \cdot u(t - 4)$

Here's the graph:



Graph of$V(t) = 12 \cdot u(t - 4)$

(ii)$V(t) = \begin{cases} 1, & 0 < t < a \\ 0, & t > a' \end{cases}$

In words, the voltage has value 1until time$t = a$. Then it is turned off.

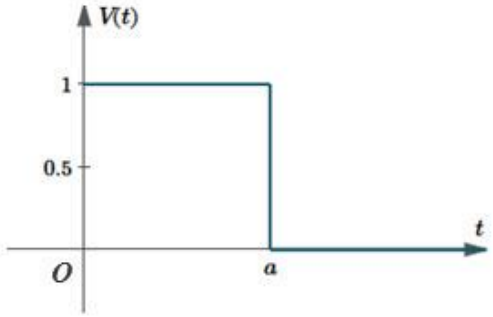
We have a "rectangular pulse" situation and need to use this formula:

$$V(t) = u(t-a) - u(t-b)$$

In our example, the pulse starts at $t = 0$, so we use $u(t)$and finishes at $t = a$ so we use $u(t-a)$.So the required function is:

$$V(t) = 1 \cdot [u(t) - u(t-a)]$$

Here is the graph



Graph of $V(t) = 1 \cdot [u(t) - u(t-a)]$, a shifted unit step function

(iii) $V(t) = \begin{cases} 0, & t < 3 \\ 2t + 8, & 3 < t < 5 \\ 0, & t > 5 \end{cases}$.

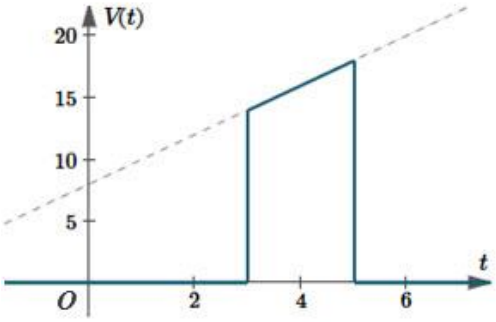
In this example, our function is $V(t) = 2t + 8$which has slope 2 and V-intercept 8.

The signal is only turned on between $t = 3$and$t = 5$. The rest of the time it is off.

So our voltage function will be:

$$V(t) = (2t + 8) \cdot [u(t-3) - u(t-5)]$$

The graph is as follows:



Graph of$V(t) = (2t + 8) \cdot [u(t-3) - u(t-5)]$. The dashed line is $V(t) = (2t + 8)$.

Now, the Laplace transform of **unit step function** is,

$$\mathcal{L}\{1.u(t-a)\} = \frac{e^{-as}}{s}$$

$$\mathcal{L}\{f(t).u(t-a)\} = e^{-as}\mathcal{L}\{f(t+a)\}$$

Example: 01

Find the Laplace transformation of $t^2u(t-3)$.

Solution: $\mathcal{L}\{t^2u(t-3)\} = e^{-3s}\mathcal{L}\{(t+3)^2\}$

$$= e^{-3s}\mathcal{L}\{t^2 + 6t + 9\}$$

$$= e^{-3s}\left[\frac{2}{s^3} + 6\frac{1}{s^2} + 9\frac{1}{s}\right].$$

Example: 02

Find the Laplace transformation of $e^{-2t}u_\pi(t)$.

Solution: $u_\pi(t) = \begin{cases} 0, & t < \pi \\ 1, & t > \pi \end{cases}$.

$$= u(t-\pi)$$

$$\mathcal{L}\{e^{-2t}u_\pi(t)\} = e^{-\pi s}\mathcal{L}\{f(t+\pi)\} \quad = e^{-\pi s}\mathcal{L}\{e^{-2(t+\pi)}\}$$

$$= e^{-\pi s}\mathcal{L}\{e^{-2t}e^{-2\pi}\} = e^{-\pi s}e^{-2\pi}\mathcal{L}\{e^{-2t}\} = e^{-\pi(s+2)}\frac{1}{s+2}.$$

Example: 03

Find the Laplace transformation of $t\,u_2(t)$.

Solution: $\mathcal{L}\{t\,u_2(t)\} = \mathcal{L}\{t\,u(t-2)\}$

$$= e^{-2s}\mathcal{L}\{t+2\}$$

$$= e^{-2s}\left(\frac{1}{s^2} + 2\frac{1}{s}\right)$$

$$= e^{-2s}\left(\frac{2}{s} + \frac{1}{s^2}\right).$$

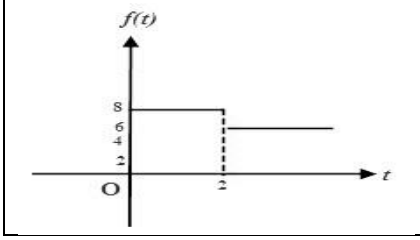
Example: 04 Given$f(t) = \begin{cases} 8, & 0 < t < 2 \\ 6, & t > 2 \end{cases}$,

(i) sketch $f(t)$,

(ii) convert $f(t)$ to unit step function and,

(iii) find the Laplace transformation of $f(t)$.

Solution:

(i) 	(ii) $f(t) = 8[u(t-0) - u(t-2)] + 6[u(t-2)]$ $= 8[u(t-0)] - 2[u(t-2)]$ (iii) $\mathcal{L}\{f(t)\} = \mathcal{L}\{8\} - 2\mathcal{L}\{u(t-2)\}$ $= \frac{8}{s} - 2.\frac{e^{-2s}}{s}.$

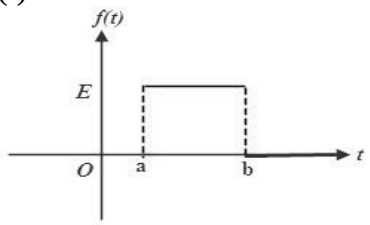
Example:05 Given$f(t) = \begin{cases} E, & a < t < b \\ 0, & t > b \end{cases}$, where E, *a* and *b* are positive constants.

(i) sketch $f(t)$,

(ii) convert $f(t)$ to unit step function and,

(iii) find the Laplace transformation of $f(t)$.

Solution:

(i) 	(ii) $f(t) = E\begin{cases} 1 & a < t < b \\ 0 & t > b \end{cases}$ $= E[u(t-a) - u(t-b)]$ (iii) $\mathcal{L}\{f(t)\} = E\mathcal{L}\{u(t-a) - u(t-b)\}$ $= E\left[\frac{e^{-as}}{s} - \frac{e^{-bs}}{s}\right].$

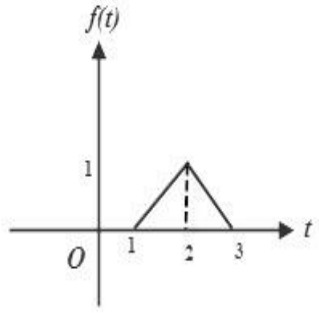
Example: 06 Given$f(t) = \begin{cases} t-1, & 1 < t < 2 \\ 3-t, & 2 < t < 3 \end{cases}$,

(i) sketch $f(t)$,

(ii) convert $f(t)$ to unit step function and,

(iii) find the Laplace transformation of $f(t)$.

Solution:

(i) 	(ii) $f(t) = (t-1)\left[u(t-1) - u(t-2)\right] + (3-t)\left[u(t-2) - u(t-3)\right]$ $= (t-1)u(t-1) - (t-1)u(t-2) + (3-t)u(t-2) - (3-t)u(t-3)$ $= (t-1)u(t-1) - 2(t-2)u(t-2) + (t-3)u(t-3)$ (iii) $\mathcal{L}\{f(t)\} = e^{-s}\mathcal{L}\{t+1-1\} - 2\,e^{-2s}\mathcal{L}\{t+2-23\} + e^{-3s}\mathcal{L}\{t+3-3\}$ $= e^{-s}\frac{1}{s^2} - 2\,e^{-2s}\frac{1}{s^2} + e^{-3s}\frac{1}{s^2}$ $= \frac{1}{s^2}(e^{-s} - 2\,e^{-2s} + e^{-3s}).$

Example: 07 Given$f(t) = \begin{cases} 9, & 0 < t < 2 \\ -6, & 2 \le t < 6 \\ 25, & 6 \le t < 9 \\ 7, & t > 9 \end{cases}$,

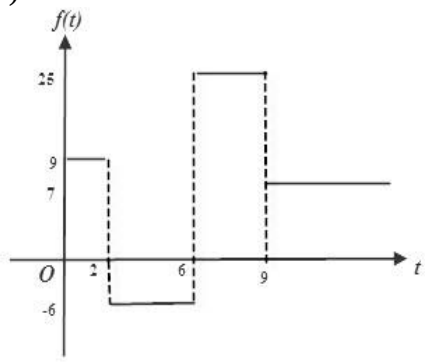
(i) sketch $f(t)$,

(ii) convert $f(t)$ to unit step function and,

(iii) find the Laplace transformation of $f(t)$.

Solution:

(i)



(ii) $f(t) = 9[u(t-0) - u(t-2)] + (-6)[u(t-2) - u(t-6)] + 25[u(t-6) - u(t-9)] + 7[u(t-9)]$

$= 9u(t-0) - 15u(t-2) + 31u(t-6) - 18u(t-9)$

(iii) $\mathcal{L}\{f(t)\} = \mathcal{L}\{9u(t-0)\} - \mathcal{L}\{15u(t-2)\} + \mathcal{L}\{31u(t-6)\} - \mathcal{L}\{18u(t-9)\}$

$= 9e^{0s}\frac{1}{s} - 15e^{-2s}\frac{1}{s} + 31e^{-6s}\frac{1}{s} - 18e^{-9s}\frac{1}{s}$

$= (9 - 15e^{-2s} + 31e^{-6s} - 18e^{-9s})\frac{1}{s}$.

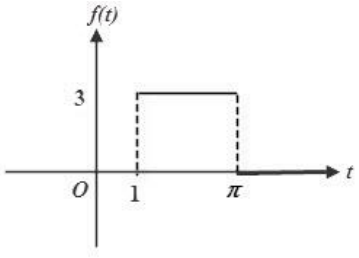
Exercises set 1.3

Sketch the function and write in terms of unit step function. Hence find the Laplace transformation.

	Functions $f(t)$	Answers
1	$t^2u_3(t)$,	$\left(\frac{2}{s^3} + \frac{6}{s^2} + \frac{9}{s}\right)e^{-3s}$,
2	$f(t) = \begin{cases} 2t, & 0 < t \le 5 \\ 0, & t > 5 \end{cases}$,	$\frac{2-2(5s+1)e^{-5s}}{s^2}$,
3	$f(t) = \begin{cases} 2, & 0 < t < 1 \\ -2, & 1 \le t < 3, \\ 1, & t \ge 3 \end{cases}$	$\frac{2}{s} - \frac{4}{s}e^{-s} + \frac{3}{s}e^{-3s}$,
4	$f(t) = \begin{cases} 2t, & 0 \le t < 6 \\ 0, & 6 \le t < 8 \\ -2, & t \ge 8 \end{cases}$,	$\frac{2}{s^2}(1 - e^{-6s} - 6se^{-6s} - se^{-8s})$,
5	$f(t) = \begin{cases} t^2, & 0 \le t < 1 \\ 0, & t \ge 1 \end{cases}$,	$-e^{-s}\left[\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s}\right]$.
6	$f(t) = \begin{cases} t-3, & 1 < t < 4 \\ 0, & \text{otherwise} \end{cases}$,	$e^{-s}\left(\frac{1}{s^2} - \frac{2}{s}\right) - e^{-4s}\left(\frac{1}{s^2} + \frac{1}{s}\right)$,

Represent the following functions in terms of unit step function and find their Laplace transformations:

1.



2.

